



AI Infrastructure Architecture Evolution

2015-2026

Silicon generations · Memory and interconnects
Conference disclosures · Systems and software

A generation-by-generation synthesis of conference disclosures and primary technical sources.
Evidence cutoff: 28 September 2026.
Complete English edition, paired with a full Simplified Chinese report.

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01 Scope and executive synthesis

Deep coverage: Graviton CPUs and Inferentia/Trainium accelerators. Full coverage: NeuronLink, UltraServer, Nitro, EFA and the Neuron software model. Period: 2015-28 September 2026; the CPU product sequence starts in 2018. FPGA instances and third-party GPUs are complementary context, not AWS-designed silicon. Reviewed milestones include re:Invent 2024/2025 materials, Graviton5 general availability in June 2026, and current Neuron architecture documentation.

AWS's architectural strategy is to make specialization deployable through a stable cloud service boundary. Graviton supplies general compute; Trainium and Inferentia supply model compute; Nitro supplies isolation and infrastructure offload; EFA supplies distributed communication. The economic unit is an instance or a tightly connected system with a supported software stack. The principal engineering question is how much specialized silicon can be exposed without forcing customers to operate a bespoke machine.

02 Numerical and availability conventions

CPU counts are per package, not the largest multi-socket EC2 instance. A Graviton vCPU represents a physical core rather than an SMT thread. Accelerator figures are per physical device and dense; cFP8, FP8 and MXFP8 are not interchangeable formats. HBM GiB and GB are retained as published. NeuronLink totals whose direction is unspecified remain quoted totals rather than invented one-way bandwidth. Service GA, preview and future roadmap are separate statuses. n/d means not established; n/a means not applicable.

03 Graviton1 and Graviton2: establishing the platform

The first Graviton A1 instances established an Arm64 deployment path in 2018. Graviton2 made that path a broad server platform through 64 Neoverse N1 cores, private L2 caches and an eight-channel memory interface. The move was more than an ISA substitution: software builds, native libraries, runtime optimization and instance sizing had to become routine. A single-thread-per-core model also changes the meaning of a vCPU when comparing against an SMT-based x86 instance.

Architectural interpretation: the migration succeeds when an application's dependency graph is portable and its working set fits the memory and cache envelope. The useful experiments are fleet-representative throughput and tail latency at a fixed service objective. Compiler-generated Arm code is necessary but insufficient: synchronization primitives, cryptography, compression and managed runtimes can determine whether silicon resources translate into service capacity.

04 Graviton3-5: vector growth and a new chiplet boundary

Graviton3 kept 64 cores while moving to Neoverse V1 and DDR5; Graviton3E is a related HPC-oriented variant rather than a separate ISA generation. Graviton4 increased the package to 96 Neoverse V2 cores and twelve DDR5 channels, with two-socket configurations exposed in selected large instances. Wider aggregate vector capability should not be confused with a fixed architectural SVE vector length: the V1 and V2 implementations organize their datapaths differently.

Graviton5 changes the partitioning: four chiplets each integrate 48 cores, memory controllers and PCIe controllers. The package has 192 cores and 192 MB L3, supports DDR5-8800 and PCIe 6, and uses Neoverse V3. The previous separate I/O and memory-controller dies disappear from this organization. The vendor quotes up to 420 GB/s between chiplets without fully specifying the directional basis; it is retained as a quoted link figure. NUMA and cache partitioning must be interpreted at the selected VM size, not inferred from the package total.

Architectural implication: placing memory controllers beside compute can reduce traffic that crosses a dedicated I/O boundary, but distributed controllers make locality and allocation policy more important. Doubling core count with twelve channels also means that per-core peak DRAM bandwidth need not rise with package bandwidth. Larger L3 can reduce traffic, but it cannot remove the streaming bandwidth limit. These two effects should be measured separately before attributing an application gain to core IPC.

Graviton generations · 1/3

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Generation / reference SKU	Codename	GA	Status	Core µarch	Cores / threads
Graviton / A1	n/d	2018	Historical shipping	Arm64	16 / 16
Graviton2	n/d	2020	Shipping / available	Neoverse N1	64 / 64
Graviton3 / 3E	n/d	2022 / 2023	Shipping / available	Neoverse V1	64 / 64
Graviton4	n/d	2024	Shipping / available	Neoverse V2	96 / 96
Graviton5	n/d	2026-06-10	Shipping / available	Neoverse V3	192 / 192

Graviton generations · 2/3

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Generation / reference SKU	Process per die	Dies / packaging	L2 per core	L3 / domain
Graviton / A1	n/d	n/d	n/d	n/d
Graviton2	n/d	n/d	1 MB	32 MB
Graviton3 / 3E	n/d	7 dies; split compute/memory/I/O	1 MB	32 MB
Graviton4	n/d	Separate controller dies	2 MB	36 MB
Graviton5	3 nm	4 × 48-core chiplets	2 MB	192 MB package; VM partitioned

Graviton generations · 3/3

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Generation / reference SKU	Memory: channels; rate; GB/s	Sockets / links	PCIe / CXL	TDP	Vector / matrix ISA
Graviton / A1	DDR4; n/d	n/d	n/d	n/d	Arm64
Graviton2	8 × DDR4; rate n/d	1	n/d	n/d	2 × 128-bit Neon
Graviton3 / 3E	8 × DDR5; rate n/d	1	n/d	n/d	2 × 256-bit SVE; BF16
Graviton4	12 × DDR5-5600; 537.6 GB/s	1-2; link n/d	n/d	n/d	4 × 128-bit SVE; SVE2
Graviton5	12 × DDR5-8800; 844.8 GB/s	NUMA 2/4 regions physical	PCIe 6; CXL n/d	n/d	4 × 128-bit SVE; Armv9.2

05 Inferentia and Trainium1: the NeuronCore foundation

Inferentia1 combines four NeuronCore-v1 units with external DRAM. It targets inference and benefits from pipelining work through cores. Inferentia2 and Trainium1 use NeuronCore-v2, HBM, separate tensor/vector/scalar engines and programmable GPSIMD resources. This shared core lineage does not make the cloud products identical: Inf2 and Trn1 expose different chip counts, networking and intended workload envelopes. The move to HBM makes larger models practical while increasing the importance of explicit data movement.

Software-managed SBUF and accumulation storage are architectural resources, not transparent caches. A kernel must arrange tensors to fit the array's contraction dimension and the scratchpad's partitioning. DMA overlap is therefore part of the algorithm. This is a useful distinction from a superficial "GPU alternative" description: the compiler and kernel author carry substantial responsibility for locality, even when a framework hides that responsibility from the model author.

06 Trainium2, Trainium3 and the Trainium4 roadmap

Trainium2 expands to eight NeuronCore-v3 units, larger on-chip storage and a 96 GiB HBM system. Logical NeuronCore configuration can combine physical resources behind a larger logical core. Dedicated collective-control cores support communication alongside computation. Trn2 UltraServers extend the directly connected resource pool beyond one conventional server, so both the physical core count and the logical execution mode matter when interpreting utilization.

Trainium3 retains eight physical cores but changes the low-precision datapath. The NKI guide explains quad-packed microscaling inputs on a physical 128×128 array, presenting a larger logical contraction dimension. The overview gives 2,517 TFLOPS for MXFP8/MXFP4 and 671 TFLOPS for dense BF16. Thus the headline low-precision improvement is not a universal BF16 improvement. Trn3 UltraServers reach 144 chips and use NeuronSwitch-v1; their aggregate improvement also includes a larger chip count.

Trainium4 remains a roadmap item in the reviewed disclosure. AWS states intended NVLink Fusion support alongside Graviton and EFA in MGX-based infrastructure. This is a future integration direction, not a reason to relabel existing NeuronLink systems as NVLink systems. Qualification must ultimately cover memory semantics, collective software, fault handling and topology as well as physical link compatibility.

07 Accelerator comparison

Inferentia and Trainium · 1/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Architecture	GA / milestone	Status	Process	Dies / packaging
Inferentia1	NeuronCore-v1	2019	Historical shipping	n/d	n/d
Trainium1	NeuronCore-v2	2022	Shipping / available	n/d	n/d
Inferentia2	NeuronCore-v2	2023	Shipping / available	n/d	n/d
Trainium2	NeuronCore-v3	2024-12	Shipping / available	n/d	n/d
Trainium3	NeuronCore-v4	2025-12	Shipping / available	3 nm	n/d
Trainium4	n/d	n/d	Roadmap	n/d	n/d

Inferentia and Trainium · 2/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Compute units	Dense BF16	Dense FP8	Dense FP6 / FP4	FP64 vector / matrix
Inferentia1	4	64 TFLOPS	n/a	n/a	n/d
Trainium1	2	190 TFLOPS	190 cFP8 TFLOPS	n/a	n/d
Inferentia2	2	190 TFLOPS	190 cFP8 TFLOPS	n/a	n/d
Trainium2	8	667 TFLOPS	1,299 TFLOPS	n/a	n/d
Trainium3	8	671 TFLOPS	2,517 MXFP8 TFLOPS	2,517 MXFP4 TFLOPS	n/d
Trainium4	n/d	n/d	n/d	n/d	n/d

Inferentia and Trainium · 3/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Memory: type; stacks; GB; TB/s	LLC / SRAM
Inferentia1	DRAM; 8 GiB; 50 GiB/s	n/d
Trainium1	HBM; 32 GiB; 820 GiB/s	48 MiB SBUF
Inferentia2	HBM; 32 GiB; 820 GiB/s	n/d
Trainium2	HBM; 96 GiB; 2.9 TB/s	224 MiB SBUF
Trainium3	HBM3e; 144 GiB; 4.9 TB/s	256 MiB SBUF
Trainium4	n/d	n/d

Inferentia and Trainium · 4/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Scale-up: links; GB/s per direction; domain	Host link	Power / cooling	Form factor
Inferentia1	NeuronLink; n/d	n/d	n/d	AWS platform
Trainium1	NeuronLink-v2; 16 chips	n/d	n/d	AWS platform
Inferentia2	NeuronLink-v2; ≤12 chips/instance	n/d	n/d	AWS platform
Trainium2	1.28 TB/s quoted; 64 chips	n/d	n/d	AWS platform
Trainium3	2.56 TB/s quoted; 144 chips	n/d	n/d	AWS platform
Trainium4	NVLink Fusion planned	n/d	n/d	AWS platform

Source reconciliation: the current Trainium3 overview reports 4.9 TB/s HBM and 16 collective cores; the NKI guide gives 4.7 TB/s and 20 CC-Cores in its introductory list. The overview is used consistently for package comparison, while the guide supports datapath discussion. Likewise, Trainium1 is rounded to 190 TFLOPS in its own guide and 191 in a later comparison. These discrepancies are retained in the evidence record rather than silently combined.

08 Nitro and EFA: infrastructure versus AI traffic

Nitro separates much of the networking, storage and management work from customer CPU execution. Its security chip, cards and minimal hypervisor form a system, not a single AI NIC. Successive Nitro versions appear across EC2 CPU and accelerator families; their network limits depend on the instance. Nitro v6 is publicly associated with the 2025 generation of network-optimized instances. The 2026 Nitro Isolation Engine adds a formally verified isolation component, which is a specific security property rather than proof that every part of the system is formally verified.

EFA presents an application communication interface backed by AWS's scalable reliable datagram transport. Multipath delivery and fast recovery address the variability of a shared datacenter network. The accelerator collective library, provider interface, EFA device and underlying fabric are separate layers. Their combined behavior determines job performance; a port line rate alone says little about small-message latency, tail behavior or communication-compute overlap. ENA front-end traffic and EFA collective traffic should also remain distinct in the system map.

Offload and network families · 1/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	GA / milestone	Status	Class
Nitro System	2017 onward	Shipping / available	Infrastructure offload / isolation
EFA	2019 onward	Shipping / available	AI / HPC network interface
NeuronSwitch-v1	2025	Shipping / available	Scale-up switch

Offload and network families · 2/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Ports × speed	SerDes	Host interface	Programmable engine
Nitro System	Instance-dependent	n/d	PCIe; n/d	n/d
EFA	Instance-dependent	n/d	n/d	n/d
NeuronSwitch-v1	n/d	n/d	n/a	n/d

Offload and network families · 3/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Embedded CPU	On-card memory	Transport / RDMA	Congestion control
Nitro System	n/d	n/d	ENA / EBS	n/d
EFA	n/d	n/d	SRD / libfabric	Multipath / recovery
NeuronSwitch-v1	n/d	n/d	NeuronLink	n/d

Offload and network families · 4/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Key offloads	Power
Nitro System	VPC, storage, management	n/d
EFA	OS-bypass communication	n/d
NeuronSwitch-v1	Device fabric	n/d

09 Systems, software and bandwidth hierarchy

Neuron provides graph compilation, runtime, libraries and framework integration. NKL exposes the hardware memory hierarchy to specialist kernel authors; logical-core configuration changes how physical resources are presented. A production migration should track supported operators, compilation cost, numerical behavior, profiling and collective overlap together. AWS FPGA instances remain a programmable acceleration option, while GPUs remain part of the same cloud portfolio; neither is evidence that AWS designed the underlying third-party silicon.

System progression · 1/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Platform	Year / milestone	CPU / accelerator count	Scale-up domain
Trn1	2022	16 Trainium1	16
Trn2 UltraServer	2024	64 Trainium2	64
Trn3 UltraServer	2025	144 Trainium3	144

System progression · 2/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Platform	Scale-out per accelerator	Rack power	Cooling
Trn1	Instance-dependent; n/d here	n/d	n/d
Trn2 UltraServer	n/d	n/d	n/d
Trn3 UltraServer	n/d	n/d	n/d

Interconnect lineage · 1/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

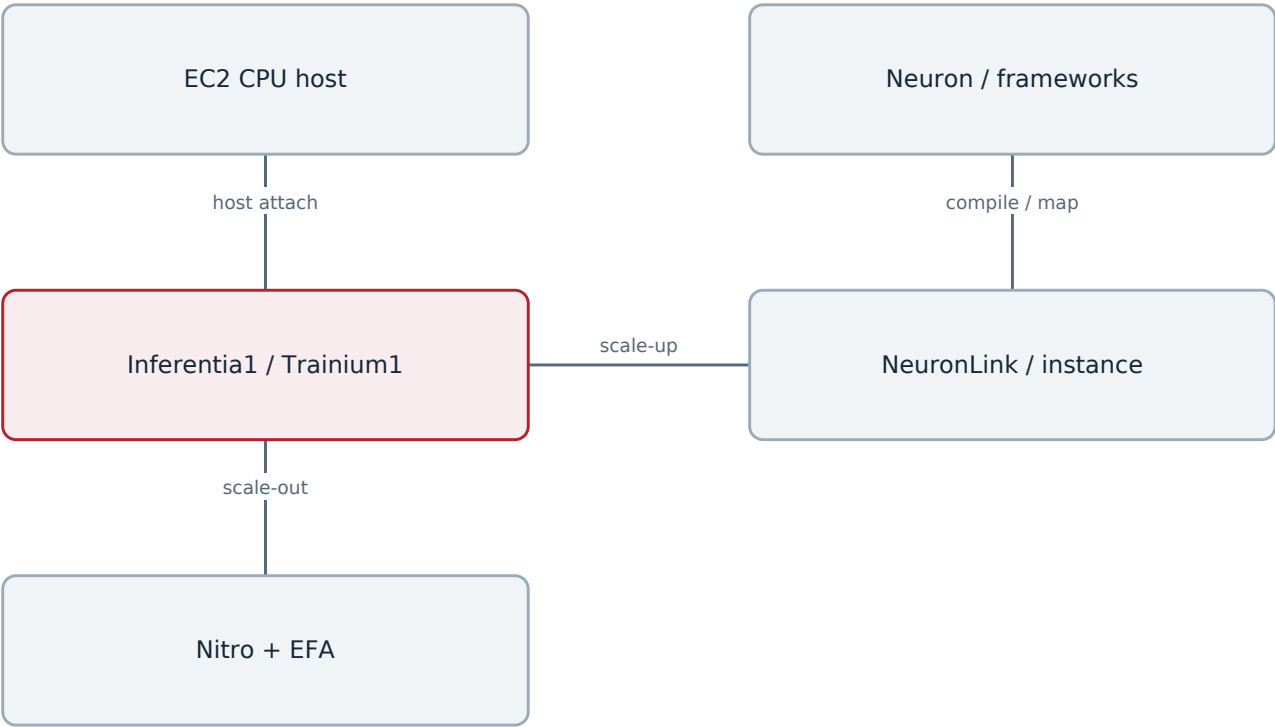
Interconnect	Year / generation	Lane rate	Lanes / link	GB/s per link per direction
NeuronLink-v2	2022	n/d	n/d	n/d
NeuronLink-v3	2024	n/d	n/d	n/d
NeuronLink-v4	2025	n/d	n/d	n/d
NVLink Fusion	n/d	n/d	n/d	n/d

Interconnect lineage · 2/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

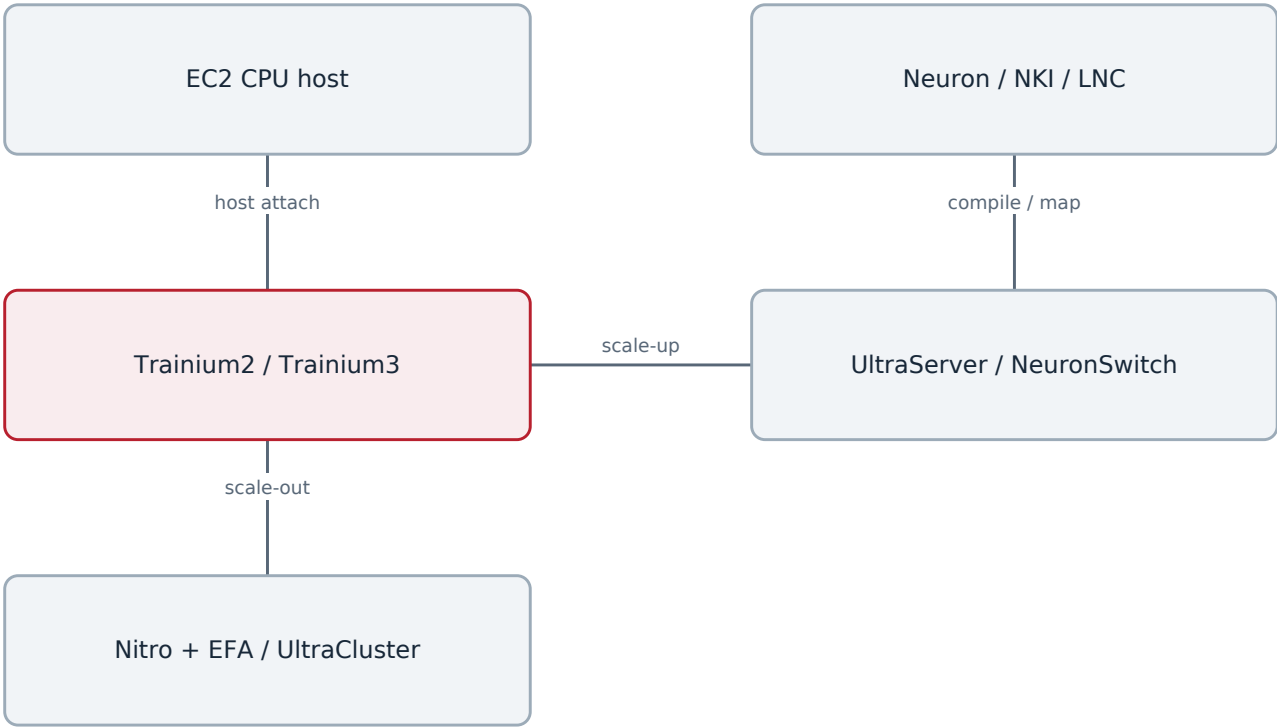
Interconnect	Links / device	Topology	Domain limit	Coherence / semantics
NeuronLink-v2	n/d	n/d	16	Device communication; not CPU coherence
NeuronLink-v3	n/d	n/d	64	Device communication; not CPU coherence
NeuronLink-v4	n/d	NeuronSwitch-v1	144	Device communication; not CPU coherence
NVLink Fusion	n/d	Planned	n/d	Device communication; not CPU coherence

2019-2022 integrated stack



Logical data and control paths. Graviton is not assumed to host every historical accelerator instance.

2024-2026 integrated stack



Logical data and control paths. Graviton is not assumed to host every historical accelerator instance.

10 Derived balance and engineering consequences

Per-device derived ratios

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Device	Memory BW / BF16, B/FLOP	BW / low-precision, B/FLOP	Capacity / BF16, B-s/FLOP	Performance / W
Trainium1	0.004634	0.004634	0.0001808	n/d
Trainium2	0.004348	0.002232	0.0001545	n/d
Trainium3	0.007303	0.001947	0.0002304	n/d

Derived CPU balance: Graviton4 provides $537.6/96 = 5.6$ GB/s per core; Graviton5 provides $844.8/192 = 4.4$ GB/s per core, a 21.4% reduction in this theoretical streaming metric. Package L3 per core rises from 0.375 MB to 1 MB, before VM partitioning. The architectural bet is that improved cores, more cache and locality compensate for less bandwidth per core on enough real workloads. A streaming workload and a large instruction-footprint service can therefore move in opposite directions.

For accelerators, Trainium3 improves memory bandwidth per BF16 FLOP substantially because BF16 compute remains nearly flat while memory bandwidth rises. Against its faster MX datapath, the bandwidth balance is tighter. This is why a procurement model should carry separate BF16 and microscaling rooflines, and why a token-throughput claim needs batch size and latency limits. System-level gains also depend on whether the model can exploit the enlarged scale-up domain without adding excessive communication or idle capacity.

11 Conference map, software timeline and naming

Verified disclosure channels · 1/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Venue	Year	Paper / talk / disclosure
re:Invent	2024	CMP207 AWS accelerated computing
re:Invent	2025	Trainium3 and Trainium4 outlook
AWS Summit	2022	The journey of silicon innovation at AWS
Amazon Science	2026	Graviton5 chiplet architecture

Verified disclosure channels · 2/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Venue	Presenter / authors	Product	Evidence type	Source
re:Invent	AWS	Trainium2 / Nitro / EFA	Architecture and systems	C01
re:Invent	AWS	Trainium3 / 4	GA / roadmap	E02 E04
AWS Summit	AWS	Graviton / Nitro	Portfolio and systems	C02
Amazon Science	Ali Saidi	Graviton5	Vendor technical disclosure; not conference paper	E01

The reviewed corpus supports stronger implementation coverage through AWS architecture guides than through a verified ISCA/MICRO/HPCA/ASPLOS or ISSCC product-paper sequence. No such sequence is invented. Hot Chips was searched, but no specific AWS generation talk is used as an implementation source in this edition. This is a disclosure-channel difference from Google, not evidence of less architectural sophistication. re:Invent materials establish deployment and system context; Neuron and Graviton guides supply the detailed programming-visible architecture.

Hardware and software alignment

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Release	Date	Hardware	Feature
NeuronCore-v1 support	2019	Inferentia1	Inference compilation / pipeline
NeuronCore-v2 support	2022-2023	Trainium1 / Inferentia2	Dynamic shapes; GPSIMD operators
NKI / LNC	2024-2026	Trainium2 / 3	Explicit kernels / logical cores

Product crosswalk

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Silicon	Service	Core / fabric
Inferentia1 / 2	Inf1 / Inf2	NeuronCore-v1 / v2
Trainium1 / 2 / 3	Trn1 / Trn2 / Trn3	NeuronCore-v2 / v3 / v4
Graviton5	M9g / related families	Neoverse V3; Nitro

Open implementation fields include package power, precise per-die process and packaging information for several accelerator generations, and fully directional NeuronLink electrical specifications. The system tables keep rack power and per-accelerator EFA bandwidth n/d when the selected configuration is not fixed. The next decision-relevant evidence is a matched workload comparison with compilation and communication costs included, rather than an extrapolation from chip peak to fleet efficiency.

Source map

- 01 Scope and executive synthesis - C01, E01, E02, E03, P01, P02, P03, P04, P07, P08
- 02 Numerical and availability conventions - P01, P02, P03, P04, P06
- 03 Graviton1 and Graviton2: establishing the platform - C01, P01
- 04 Graviton3-5: vector growth and a new chiplet boundary - E01, E03, E05, E06, E07, P01
- 05 Inferentia and Trainium1: the NeuronCore foundation - P02, P05, P06, P09
- 06 Trainium2, Trainium3 and the Trainium4 roadmap - C01, E02, P03, P04, P09
- 07 Accelerator comparison - E02, E04, E08, E09, P02, P03, P04, P05, P06, P09
- 08 Nitro and EFA: infrastructure versus AI traffic - E01, E02, P07, P08, P10, P11
- 09 Systems, software and bandwidth hierarchy - C01, E02, P02, P03, P04, P09
- 10 Derived balance and engineering consequences - E01, E02, E05, P01, P02, P03, P04, P09
- 11 Conference map, software timeline and naming - C01, C02, E01, E02, E03, E04, P01, P02, P03, P04, P05, P06, P08, P09

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[P02] Trainium architecture

<https://awsdocs-neuron.readthedocs-hosted.com/en/latest/about-neuron/arch/neuron-hardware/trainium.html>

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[P03] Trainium2 architecture

<https://awsdocs-neuron.readthedocs-hosted.com/en/latest/about-neuron/arch/neuron-hardware/trainium2.html>

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[P04] Trainium3 architecture

<https://awsdocs-neuron.readthedocs-hosted.com/en/latest/about-neuron/arch/neuron-hardware/trainium3.html>

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[P05] Inferentia architecture

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[P06] Inferentia2 architecture

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[P07] Security design of the AWS Nitro System

<https://docs.aws.amazon.com/whitepapers/latest/security-design-of-aws-nitro-system/security-design-of-aws-nitro-system.html>

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[P08] Elastic Fabric Adapter

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/efa.html>

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[P09] Trainium3 NKI architecture guide

https://awsdocs-neuron.readthedocs-hosted.com/en/latest/nki/guides/architecture/trainium3_arch.html

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[P10] EC2 Nitro versions

<https://docs.aws.amazon.com/ec2/latest/instancetypes/ec2-nitro-instances.html>

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[P11] SRD transport

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Launches, ecosystem and deployment

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[E03] Graviton5 M9g general availability

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[E04] Trainium3 general availability

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[E05] Graviton4 memory architecture

<https://aws.amazon.com/blogs/aws/join-the-preview-for-new-memory-optimized-aws-graviton4-powered-amazon-ec2-instances-s-r8g/>

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[E06] Graviton3 C7g GA

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[E07] Graviton A1 launch

<https://aws.amazon.com/blogs/aws/new-ec2-instances-a1-powered-by-arm-based-aws-graviton-processors/>

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[E08] Inferentia Inf1 launch

<https://aws.amazon.com/blogs/aws/amazon-ec2-inf1-instances-with-aws-inferentia-chips-for-high-performance-cost-effective-machine-learning-inference/>

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[E09] Inf2 instances

<https://aws.amazon.com/ec2/instance-types/inf2/>

Indexed record or searchable official page; product details cross-checked with the associated technical document.